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“NO LIMITS, JUST LANGUAGE” FUN, FUTURE, AND LINGUISTICS

Study of language has been close to us since the ancient Indian grammarian Panini worked out to scientifically understand Sanskrit across time and tradition. Today, linguistics boomerangs between not just tradition and time, but modern concerns of technology and cognition.

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Linguistics has a confession to make. It is just like you. It tells you that it is as emotional, messy, dynamic, strategic and perfectly imperfect as any human being. Whatever you are, linguistics becomes. Why? Because linguistics watches you. It watches you with the same clinical precision of a surgeon. It watches you speak and communicate, how your speech changes over time, how you chaotically mix languages, make up new words formally or for funsies, follow rules and patterns, rebel even, and at the end of it all, how these stem ardently from within you and not apart. So yes, it is just like you and will not go away.

Creative and logical
IIIT- Hyderabad's Linguistics professor Parameswari Krishnamurthy says that what constantly excites her most about the discipline is that "it's both creative and

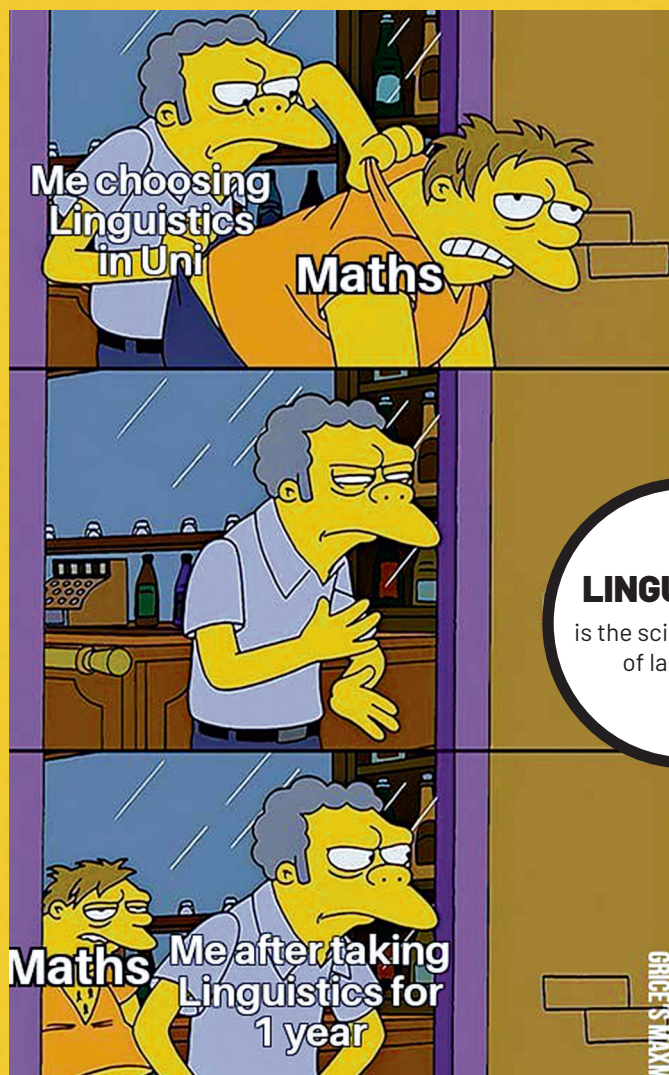


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logical". It's like mathematics and literature marrying each other when on a normal day, people regard them as two sides of a coin that do not see eye to eye. "We use language everyday and you unconsciously know that there are certain facts about the language which you don't know why but you still use it", she describes. Starting to uncover those facts or truth is what happens when you get into linguistics for the first time.

We use language to communicate and so "anything that happens in human lives is reflected in language." For instance, with internet slang, a lot of our active speech vocabulary includes its usage and thus, reflects how the internet has become so integral to our lives.

All these languages are inherently creative, so bringing a scientific lens to it unveils patterns and problems that spike up your curiosity.

You're my missing puzzle piece

This year, IIIT Hyderabad-backed Indian team won multiple medals at the International Linguistics Olympiad conducted in Romania. Krishnamurthy, who was part of it and led the team,



WHAT DO LINGUISTS STUDY?

Instead of looking at things going around with languages, linguistics looks directly at it. It questions why water is called as water in English and pani in Hindi, what pattern of sounds make up the word, how sentences are structured in different languages, how dialects can change those structures, how slangs change with social groups and with age, why we say things the way we say it, and the list just goes on.

Formally, the elements can be classified as word order (syntax), grammar, semantics (meaning), phonetics (sounds), culture, pragmatics (relationship between language and its speaker), and so on.

Subdisciplines inside the field include sociolinguistics (study of relationship between language and society), psycholinguistics (study of interrelation between linguistics factors and the brain), computational linguistics (uses computer science and maths to model languages) and more.

LINGUISTICS

is the scientific study of language.

explained how these patterns and problems are truly like puzzles. In a linguistic olympiad, it is just full puzzle mode. Unlike science olympiads with its theorems, Krishnamurthy says “there is some reality” here because real language data is given to you with which you try to figure out patterns.

In fact, Krishnamurthy says that it is like machine learning where the machine gets a small amount of data and it learns the pattern. Except we are creatively doing it. It is also important to note here that just as human life is not absolute and has many exceptions, language too has patterns that work and nuances that do not go with the pattern at all. For this reason, linguistics is “very, very interesting”, exclaims Krishnamurthy.

Nothing’s fully Greek in linguistics

Linguistics olympiads can give you a language you’ve



As long as humans live, the study of language lives.

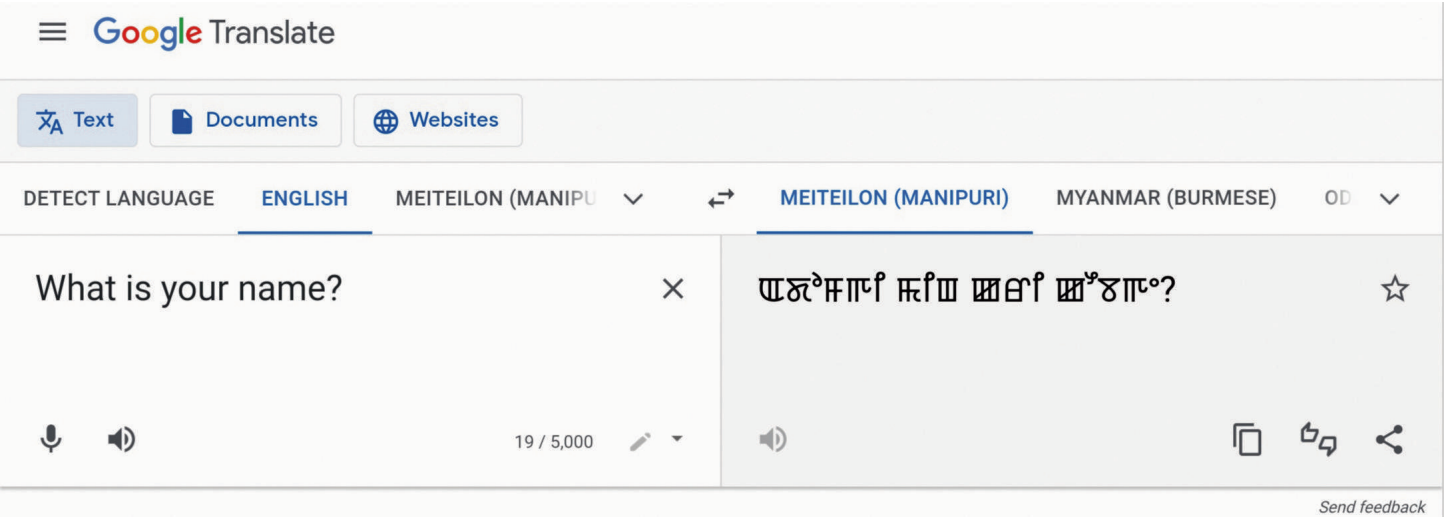
Prof. Parameshwari Krishnamurthy

never heard in your life ever, give you example data and a problem, then prompt you to figure out how that language works. “The rest of the problem is to find out the logic behind it”.

A student has to hypothesise, figure out and try patterns, compare and contrast, and eventually reach the answer. “For example, you are given three words in a language and their meaning, and you notice there is some kind of pattern, say, a plural form which is repeated multiple times. Then you get another example from which you can predict what the plural form will be. So, here basically, you are discovering grammar from clues”, says Krishnamurthy.

Conlang

This is also how Conlangs (constructed languages) work. In films like Lord of the Rings, Avatar, etc. fictional languages called conlangs are



created which are often inspired from linguistic factors and patterns from real languages. Linguists can make up a legit-sounding, fake language that is simply “not just putting words together”, says Krishnamurthy. This is one of the fun things that come out of learning linguistics. Conlangs have proper grammar, structure, and even social and dialectal differences, if the maker is nerdy enough to fine tune the

richness of it.

Tech way of life

When the nature of our language use differed and changed with the inventions of the computer, and later internet, linguistics stayed and gave us new perspectives. Computational linguistics began as a new field trying to understand how technology engaged with language. Often, when you want to know what a

sentence in a foreign language means, you pull out Google Translate to instantly translate it, right? Well, that Google Translate is a prime example of computational linguistics in action.

Into the future

Today, AI is all the rage. “Large language models (LLMs) interact with human beings. You ask questions, it tries to give you answers, summarise documents,

customise information and there is some level of translation involved. All of these are nothing but working with language”, Krishnamurthy says.

Even these AI models are changing by the second and “becoming very strong in terms of their computational power”.

Ultimately “Language is not like a calculator” and technology “cannot replace people who speak the language”, she says. She reminds us that as much as we should leverage language technologies, we should remember that they make mistakes, misunderstand cultural differences, and aren’t creative because “creativity is human-centric”. Right now there is a gap somewhere in that process which Krishnamurthy believes will “organically be filled in”.

WANT TO TRY?

If you want to try your hand, the **Panini Linguistics Olympiad (PLO)** is what you’re looking for. PLO is actually India’s selection test for International Linguistics Olympiads. It is open to all students enrolled in a school in India and are in classes from 6 through 12. You don’t need to have any language or linguistics certificate to participate.

After a series of tests on logic, problem-solving, and analysis of linguistic elements, historical and cultural factors, eight students will eventually be selected to represent Team India.